

TrES-3b

R-2602

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_260707 · created 2026-08-28T00:48:51+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2461228.8665 +/- 0.0019 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.092 +/- 0.056
Transit depth (Rp/Rs)^2	0.86 +/- 1.04 [%]
Orbital Inclination (inc)	86.73 +/- 2.9
Airmass coefficient 1 (a1)	2.99 +/- 0.95
Airmass coefficient 2 (a2)	-0.86 +/- 0.25
Scatter in the residuals of the lightcurve fit is	33.7 %
Best Comparison Star	None
Optimal Aperture	5.1
Optimal Annulus	10.59
Transit Duration (day)	0.07 +/- 0.015

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.092 $\pm$ 0.056 vs 0.16309 (deviation 1.27 σ)
Duration fit vs published	0.07 d vs 0.0567 d (deviation 0.89 σ)
Mid-time O-C	-2.4 min
Depth SNR	1.6
Residual scatter	33.7 %

Light curve

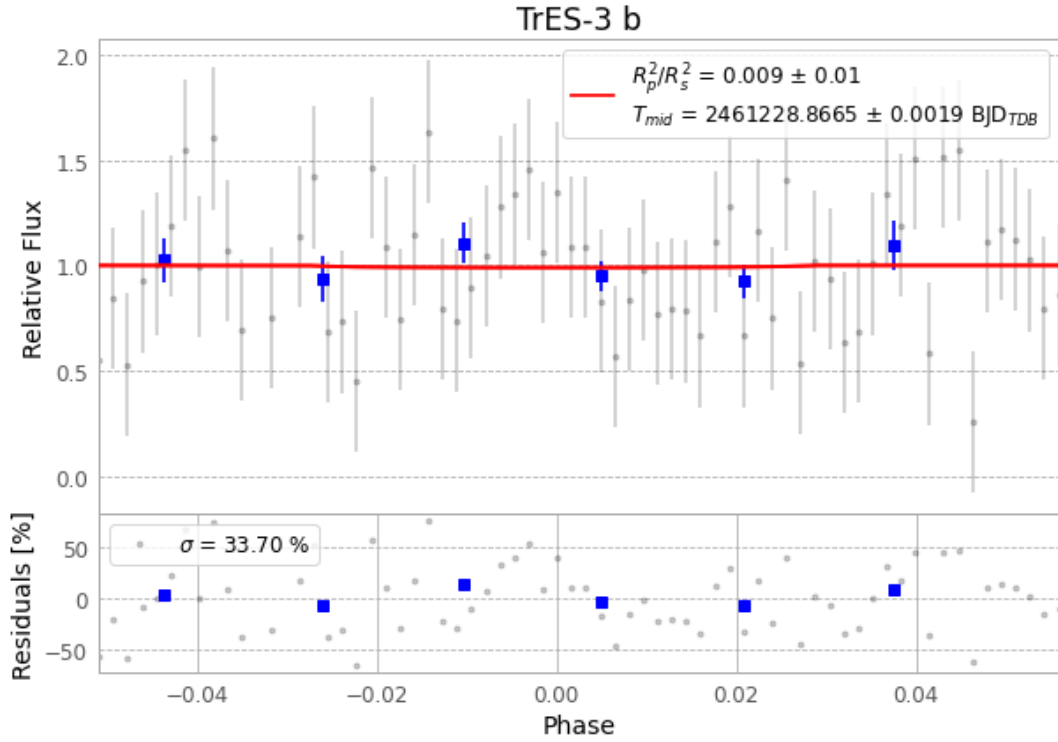


Figure 1 — FinalLightCurve_TrES-3 b_2026-07-07.png (SHA-256 3f2b185f0b64ac21...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [284, 253], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 3576e316f4044287...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

138 FITS frames (60 MB), TRES-3260707070044.FITS.gz → TRES-3260707102716.FITS.gz. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-260707.log (3eb14a0e94ac...), FinalLightCurve_TrES-3 b_2026-07-07.png (3f2b185f0b64...), FinalLightCurve_TrES-3 b_2026-07-07.csv (e878bf29b6a7...)

Compute resources

Wall time 210.8 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-28 00:48Z.